Design and Development of IoT-Based Smart Maldive Fish Processing System

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Project Proposal Report

Ramesh J N G A – IT22341686

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Ramesh J N G A	IT22341686	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
MR. RAVI SUPUNYA	11/03/2026	

ABSTRACT

Maldive fish processing involves complex interactions among mechanical systems, water-level management, salt regulation, humidity control, airflow balance, and thermal conditions. However, current processing facilities largely depend on manual supervision and isolated control mechanisms, resulting in inconsistent mechanical efficiency, uneven salinity distribution, fluctuating humidity levels, excessive energy consumption, and variable product quality.

The key research gap lies in the absence of a unified, AI-driven system that integrates mechanical, hydrological, chemical, and visual quality parameters into a single adaptive optimization framework. Existing solutions monitor these variables independently without enabling intelligent cross-parameter optimization or predictive performance control.

This research proposes an Integrated Performance Evaluation and AI-Controlled Quality Optimization framework tailored specifically for Maldive fish processing. The system incorporates calibrated mechanical sensors (RPM, torque, motor load), water-level and salinity sensors, humidity and airflow monitors, energy meters, and computer vision modules for surface quality assessment. An IoT-based real-time data acquisition pipeline will collect and synchronize multi-source sensor data. Multi-sensor fusion techniques will be applied to generate a comprehensive operational profile.

A unified AI model combining machine learning and reinforcement learning algorithms will dynamically optimize airflow, humidity stabilization, hot water levels, salt distribution, and mechanical efficiency. The system will enable predictive maintenance, adaptive process control, early anomaly detection, and automated quality assurance through continuous feedback loops.

Expected outcomes include improved mechanical reliability, optimized salt uniformity, stabilized environmental conditions, reduced energy consumption, minimized material waste, and enhanced product consistency. Socially, the solution promotes sustainable fish processing, reduced resource wastage, and higher food safety standards. Commercially, it increases productivity, lowers operational costs, and strengthens export competitiveness through standardized quality assurance.

Keywords: AI-driven optimization, multi-sensor fusion, mechanical efficiency monitoring, real-time salinity regulation, reinforcement learning control systems.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ANOVA	Analysis of Variance
API	Application Programming Interface
CNN	Convolutional Neural Network
CPS	Cyber-Physical System Sinha
ESP32	Espressif Systems 32-bit Microcontroller
GDP	Gross Domestic Product
IDE	Integrated Development Environment
IEEE	Institute of Electrical and Electronics Engineers
IoT	Internet of Things
PLC	Programmable Logic Controller
UI	User Interface
UX	User Experience
PID	Proportional-Integral-Derivative Controller

INTRODUCTION

1.1 Background and Context

Real-World Problem

The Maldive fish processing industry is a vital traditional sector in Sri Lanka and the Maldives, contributing significantly to fisheries-based livelihoods, small- and medium-scale enterprises (SMEs), and seafood export markets. Maldive fish is produced through a sequence of boiling, salting, smoking, and drying processes. However, most processing facilities continue to rely on conventional, manually controlled methods without real-time monitoring or intelligent automation.

In typical processing operations, fish are first boiled in hot water tanks, then transferred to saltwater for further boiling and salt absorption, followed by drying. These stages are manually supervised, and key operational parameters such as temperature, salinity concentration, humidity, airflow, and water levels are adjusted based on operator experience rather than precise digital measurements. This leads to inconsistent salt penetration, uneven drying, mechanical inefficiencies, and increased energy consumption.

The absence of integrated monitoring and predictive control systems results in variability in product quality, higher spoilage rates, and limited scalability.

Who Is Affected and Why It Is Important

The inefficiencies in Maldive fish processing affect multiple stakeholders:

- **Fishermen** – Post-harvest losses reduce income stability.
- **Small- and Medium-Scale Processors** – Inconsistent production limits competitiveness and profitability.
- **Exporters** – Variability in product standards affects compliance with international quality requirements.
- **Consumers** – Irregular taste, texture, and shelf-life characteristics.
- **National Economies** – Fisheries contribute substantially to employment and foreign exchange earnings in coastal regions.

Improving process efficiency and quality consistency is therefore essential for economic sustainability, food security, and industrial modernization.

Statistical Evidence (Local and Global Context)

Globally, the fisheries and aquaculture sector supports more than 60 million people directly and contributes significantly to global food supply and trade. In developing countries, post-harvest losses

in seafood processing are estimated to range between 20% and 30%, mainly due to inadequate preservation, poor monitoring systems, and inefficient processing techniques.

In Sri Lanka and the Maldives, fisheries remain a key contributor to GDP and rural employment, highlighting the need for technological advancement to reduce losses and improve export quality standards.

Analysis of Existing Solutions and Their Limitations

Current Maldive fish processing systems are largely:

- Manually operated
- Based on isolated mechanical equipment
- Lacking centralized monitoring platforms
- Dependent on visual inspection for quality control

Although modern food industries have adopted IoT, AI, and automated control systems, their application in Maldive fish processing is minimal. Existing technological implementations are fragmented and focus only on individual aspects such as drying or salinity measurement, without integrating mechanical efficiency, environmental conditions, and quality inspection into a unified intelligent framework.
As a result, process optimization remains reactive rather than predictive.

Ideas for the proposed solution and its parts

This research proposes an **Integrated IoT-Based Smart Maldive Fish Processing System** that includes:

1. **Multi-Sensor Data Acquisition System**
    - RPM and torque sensors for mechanical monitoring
    - Temperature and humidity sensors
    - Water-level sensors
    - Conductivity and salinity sensors
    - Energy consumption meters
2. **AI-Driven Predictive Analytics**
    - Mechanical efficiency prediction
    - Salinity uniformity modeling
    - Moisture and drying optimization
3. **Computer Vision-Based Quality Inspection**
    - Surface defect detection
    - Color and texture analysis
4. **Reinforcement Learning-Based Adaptive Control**
    - Dynamic airflow regulation
    - Automated water-level management
    - Real-time salt concentration adjustment
5. **Centralized Dashboard and Cloud Platform**
    - Real-time monitoring

    - Data visualization
    - Automated alerts and performance reports
- This integrated approach transitions the industry from manual supervision to a data-driven, autonomous, and optimized processing system.

Relevant SDG alignment

The proposed research aligns with the following United Nations Sustainable Development Goals (SDGs):

- SDG 2 – Zero Hunger: Reducing post-harvest losses and improving food preservation.
- SDG 8 – Decent Work and Economic Growth: Enhancing productivity and supporting SME modernization.
- SDG 9 – Industry, Innovation, and Infrastructure: Promoting smart manufacturing and technological innovation.
- SDG 12 – Responsible Consumption and Production: Improving energy efficiency and minimizing resource waste.

Research cluster alignment

This research aligns with the following research clusters:

- Artificial Intelligence and Machine Learning
- Internet of Things (IoT) and Embedded Systems
- Industrial Automation and Smart Manufacturing
- Sustainable Food Processing Technologies

1.2 Literature Review

This section reviews recent scholarly research related to Cyber-Physical Systems (CPS), IoT-enabled industrial automation, artificial intelligence-based quality monitoring, multi-sensor data fusion, reinforcement learning control, and smart food processing technologies. The review also evaluates current research approaches and identifies limitations that motivate the proposed system.

IoT-Based Smart Monitoring in Food Processing

The integration of Internet of Things (IoT) technologies into food processing industries has significantly improved monitoring and process control capabilities. Distributed sensor networks enable continuous measurement of parameters such as temperature, humidity, airflow, and machine performance. These systems support real-time monitoring and early detection of abnormal conditions in industrial processes. Studies show that IoT-enabled monitoring platforms can significantly reduce energy consumption while improving operational efficiency in food drying systems **[1]**.

Machine Learning for Quality Prediction

Machine learning models have been widely used to predict food product quality and process efficiency. Algorithms such as Support Vector Machines, Random Forest, and Artificial Neural Networks are capable of modeling complex relationships between environmental parameters and product characteristics. These models have demonstrated strong predictive performance when estimating moisture levels, salinity distribution, and product consistency in food processing environments. Experimental studies indicate that machine learning models can achieve prediction accuracies exceeding 90% when evaluating food quality parameters **[2]**.

Multi-Sensor Data Fusion in Industrial Monitoring

Industrial processes often involve multiple interconnected variables that must be monitored simultaneously. Multi-sensor data fusion techniques combine heterogeneous sensor inputs such as vibration signals, environmental measurements, and chemical indicators to provide a comprehensive operational view. This integrated approach improves the reliability of monitoring systems and enhances anomaly detection capabilities. Research has demonstrated that multi-sensor fusion architectures can improve anomaly detection accuracy by approximately 15% compared with single-sensor monitoring approaches **[3]**.

Machine Learning for Drying Process Optimization

Drying plays a critical role in seafood preservation and strongly influences product shelf life and quality. Machine learning techniques have been applied to model drying dynamics and optimize environmental conditions such as airflow, humidity, and temperature. Predictive models help determine optimal drying durations while minimizing energy consumption. Experimental studies have reported that machine learning-based optimization methods can reduce drying time by approximately 22% while maintaining acceptable quality standards **[4]**.

Computer Vision for Fish Quality Inspection

Computer vision technologies have been increasingly used for automated food quality assessment. Convolutional Neural Networks (CNNs) can analyze visual characteristics such as color variation, surface texture, and structural defects to classify food products into different quality categories. Automated vision systems reduce subjectivity in manual inspection and provide consistent product grading. Deep learning-based fish quality classification systems have achieved accuracy levels of up to 95% when identifying visual defects and inconsistencies in processed fish products **[5]**.

Reinforcement Learning for Adaptive Process Control

Traditional industrial controllers often rely on predefined parameters that cannot adapt effectively to dynamic environmental conditions. Reinforcement learning provides a promising solution for adaptive control by enabling systems to learn optimal control strategies through interaction with the environment. Reinforcement learning algorithms can dynamically adjust operational parameters such as temperature, airflow, and heating intensity to maintain process stability. Studies show that reinforcement learning-based control systems can achieve approximately 30% better temperature stability compared with conventional PID controllers in industrial thermal processes **[6]**.

Predictive Maintenance Using Machine Learning

Predictive maintenance techniques analyze operational data to detect early signs of equipment failure. Sensor data such as vibration patterns, motor speed, and power consumption can be used to train machine learning models that predict potential mechanical failures. These systems enable maintenance actions before critical failures occur, reducing downtime and operational costs. Research has shown that vibration-based predictive maintenance models can achieve up to 85% accuracy in predicting equipment failures several hours in advance **[7]**.

Cyber-Physical Systems in Smart Manufacturing

Cyber-Physical Systems integrate sensing, computation, networking, and control into a unified intelligent infrastructure. CPS architectures enable real-time interaction between physical machinery and digital monitoring platforms. These systems are considered fundamental components of Industry 4.0 manufacturing environments. The five-layer CPS architecture has been widely adopted as a reference model for developing intelligent industrial automation systems **[8]**.

AI-Driven Energy Optimization in Food Processing

Energy consumption is a major concern in food processing industries, particularly in heating and drying processes. Artificial intelligence-based optimization models can analyze energy usage patterns and dynamically adjust system parameters to improve efficiency. Research has shown that AI-based energy optimization strategies can reduce energy consumption in fish processing systems by up to 25% while maintaining stable operational conditions **[9]**.

Cloud-Based IoT Platforms for Food Supply Chains

Cloud computing technologies are increasingly used to support IoT-based monitoring and traceability systems in food supply chains. Cloud platforms enable centralized storage, real-time analytics, and remote monitoring of production systems. These platforms facilitate data sharing across different processing units and improve transparency in seafood production environments **[10]**.

1.3 Research Gap

What is Missing in Existing Research

Although significant advancements have been made in Cyber-Physical Systems (CPS), Internet of Things (IoT) monitoring, Artificial Intelligence (AI), and intelligent control mechanisms, several critical gaps remain in the context of industrial food processing, particularly in traditional seafood processing systems.

First, existing research largely focuses on isolated components such as temperature monitoring, drying optimization, salinity detection, or defect identification. There is no fully integrated end-to-end Cyber-Physical System that combines mechanical efficiency monitoring, water-level control, salinity regulation, environmental stabilization, energy optimization, and computer vision-based quality inspection into a unified intelligent framework.

Second, most AI-based models are developed and validated in laboratory environments using historical datasets. These models are rarely embedded within real-time industrial control loops for continuous adaptive optimization.

Third, reinforcement learning-based adaptive control systems remain mostly theoretical or experimentally limited, with very few implementations in real-world food processing environments.

Finally, there is limited research specifically tailored to traditional Maldive fish processing systems, which still rely heavily on manual supervision and experience-based parameter adjustment.

Why Current Approaches Are Insufficient

Current industrial approaches are insufficient due to the following reasons:

1. Fragmented Architecture – Sensors, controllers, and AI modules often operate independently without holistic system integration.

2. Static Setpoint Control – Most industrial control systems rely on predefined setpoints rather than adaptive, learning-based optimization.
3. Limited Multi-Variable Optimization – Existing systems typically optimize a single parameter (e.g., temperature or humidity) instead of simultaneously managing mechanical performance, salinity distribution, airflow, and water levels.
4. Reactive Maintenance – Many facilities adopt reactive maintenance strategies rather than predictive, AI-driven maintenance models.
5. Scalability Constraints – Advanced automation systems are often expensive and not designed for small- and medium-scale enterprises (SMEs).

As a result, traditional processing facilities continue to experience inconsistent product quality, uneven salinity penetration, energy inefficiency, mechanical wear, and higher post-processing losses.

Justification of Novelty and Creativity

The proposed research introduces a novel and creative contribution by developing a fully integrated AI-controlled Cyber-Physical System tailored specifically for Maldive fish processing.

The novelty lies in:

• Integrating mechanical, hydrological, chemical, environmental, and visual quality parameters into a single unified CPS architecture.
• Implementing reinforcement learning-based adaptive control for real-time optimization of water levels, salinity concentration, airflow, and temperature.
• Applying multi-sensor fusion techniques to generate a holistic process intelligence model.
• Combining predictive maintenance analytics with quality optimization within one intelligent platform.
• Designing a scalable and cost-effective solution suitable for SME-level deployment.

Unlike previous research that addresses isolated subsystems, this study proposes a continuous self-optimizing ecosystem capable of real-time decision-making and autonomous process improvement.

Evidence-Based Argument

Global studies report that post-harvest losses in seafood processing can range between 20% and 30% in developing regions due to inadequate monitoring, inconsistent preservation methods, and inefficient resource utilization. Research in smart manufacturing demonstrates that IoT-enabled CPS frameworks significantly improve energy efficiency, operational stability, and product consistency.

However, the literature consistently indicates that:

• AI models are often not integrated into live industrial control systems.
• CPS implementations are concentrated in high-tech manufacturing sectors rather than traditional seafood processing.
• Multi-variable reinforcement learning control in food processing environments is still at an early stage of development.

These evidence-based observations clearly justify the necessity for an integrated, AI-driven, real-time optimization framework specifically designed for Maldive fish processing.

Comparison of Existing Research Methods and the Proposed Method

Table 1- Comparison of Existing Research Methods and the Proposed Method

Technologies and Methods	**[1]**	**[2]**	**[3]**	**[4]**	**[5]**	**[6]**	**Proposed Method**
IoT-based real-time sensor monitoring	✔	✔	✔	✘	✘	✔	✔
Multi-sensor data fusion	✘	✘	✔	✘	✘	✘	✔
AI / Machine Learning prediction models	✘	✔	✔	✔	✘	✔	✔
Computer vision-based quality inspection	✘	✘	✘	✘	✔	✘	✔
Reinforcement learning adaptive control	✘	✘	✘	✘	✘	✔	✔
Mechanical efficiency monitoring	✘	✘	✘	✘	✘	✔	✔
Salinity and water-level monitoring	✔	✘	✘	✔	✘	✘	✔
Integrated Cyber-Physical System architecture	✘	✘	✔	✘	✘	✘	✔

Conclusion

The primary research gap lies in the absence of a unified, AI-controlled Cyber-Physical System that holistically integrates mechanical efficiency monitoring, water-level regulation, salinity management, environmental stabilization, predictive maintenance, and intelligent quality inspection into a single adaptive platform.

Addressing this gap will contribute to industrial modernization, improved product consistency, enhanced energy efficiency, and sustainable seafood processing innovation.

2. OBJECTIVE

2.1 Main Objective

To design, develop, and experimentally validate an integrated AI-controlled Cyber-Physical System (CPS) that improves mechanical efficiency, water-level regulation, salinity control, environmental stability, and overall product quality in Maldive fish processing by at least 20% compared to conventional manual processing methods.

2.2 Specific Objectives

1. To design and implement a multi-sensor data acquisition system capable of continuously monitoring temperature, humidity, water level, salinity concentration, airflow, and mechanical performance parameters in real time.
2. To develop a multi-variable optimization model that integrates mechanical efficiency, environmental conditions, and chemical parameters for improved process stability.
3. To implement a reinforcement learning-based adaptive control algorithm for dynamic regulation of water levels, salt concentration, airflow, and heating intensity.
4. To design and integrate a computer vision-based quality inspection module for detecting texture uniformity, surface defects, and color consistency of processed fish.
5. To develop a predictive maintenance model that analyzes motor vibration, power consumption, and operational load to reduce unexpected equipment failures by at least 15%.
6. To evaluate the overall system performance in terms of energy efficiency, product consistency, processing time reduction, and post-processing loss minimization under real operating conditions.

3. METHODOLOGY

Research Approach

This study adopts a Design Science Research (DSR) approach combined with an Experimental research methodology. The Design Science approach is used to design and develop the integrated AI-controlled Cyber-Physical System (CPS) artifact, while the Experimental method is used to evaluate and validate system performance under real processing conditions.

The research process includes problem identification, system architecture design, prototype development, implementation, experimental validation, and performance evaluation.

Technical Methods and Algorithms

The proposed system integrates the following technical methods and algorithms:

• Multi-Sensor Data Acquisition using temperature, humidity, salinity, water-level, airflow, vibration, and current sensors.

• Signal preprocessing and noise filtering using digital filtering techniques.

• Multi-variable optimization using Machine Learning regression models.

• Reinforcement Learning (RL)-based adaptive control for dynamic regulation of salinity, water level, and heating intensity.

• Computer Vision algorithms (CNN-based models) for surface defect detection and quality classification.

• Predictive maintenance modeling using time-series analysis and anomaly detection techniques.

• Multi-sensor data fusion using feature-level integration methods.

These algorithms enable real-time monitoring, prediction, and adaptive process optimization.

Tools and Platforms with Justification

The following tools and platforms will be used:

• Microcontroller/Embedded Platform (e.g., ESP32 or Arduino) – for real-time sensor integration and control.

• Raspberry Pi or Edge Computing Device – for local AI processing and data aggregation.

• Python Programming Language – for machine learning model development and data analysis.

• TensorFlow/ PyTorch – for deep learning and computer vision implementation.

• MATLAB/Simulink – for control system modeling and simulation.

• Cloud IoT Platform (e.g., Things Board or AWS IoT) – for remote monitoring and data storage.

• SQL/NoSQL Database – for structured data management.

These platforms are selected based on scalability, cost-effectiveness, compatibility with IoT infrastructure, and support for AI model deployment.

Data Collection and Ethical Considerations

Data will be collected from experimental Maldive fish processing batches under controlled operating conditions. The following data categories will be recorded:

• Mechanical data (motor speed, vibration, power consumption)

• Environmental data (temperature, humidity, airflow)

• Chemical data (salinity concentration)

• Hydrological data (water level measurements)

• Visual quality data (image-based inspection results)

Ethical considerations include:

• Ensuring no harm to workers during system deployment and testing.

• Maintaining data confidentiality and secure storage of operational data.

• Using collected data solely for research and performance evaluation purposes.

• Ensuring compliance with institutional research guidelines.

No human personal data will be collected in this study.

Validation Strategy and Performance Metrics

The system will be validated using comparative experimental analysis between conventional manual processing and the proposed AI-controlled CPS system.

Performance metrics include:

• Percentage improvement in product quality consistency.

• Reduction in salinity variance across batches.

• Improvement in mechanical efficiency (%).

• Reduction in energy consumption (kWh per batch).

• Reduction in processing time.

• Decrease in equipment failure rate (%).

• Overall reduction in post-processing losses (%).

Statistical analysis (ANOVA and paired t-tests) will be used to determine the significance of improvements. System robustness, scalability, and real-time response performance will also be evaluated.

This methodology ensures a systematic, measurable, and technically rigorous development and validation of the proposed intelligent processing framework.

TECHNOLOGIES AND PLATFORMS

Hardware Technologies

Raspberry Pi 4 – Used as the primary edge computing device for real-time AI processing, sensor data aggregation, and camera-based defect detection. It provides sufficient computational power for running lightweight deep learning models while remaining cost-effective and suitable for SME-level deployment.

AI & Machine Learning Technologies

Programming Language

Python – Selected as the primary programming language for AI model development, sensor data processing, backend integration, and control algorithm implementation. Python offers extensive libraries, strong community support, and seamless integration with IoT and machine learning frameworks.

Deep Learning Framework

PyTorch – Used as the main deep learning framework for computer vision tasks, defect detection, predictive modeling, and reinforcement learning implementation. PyTorch is chosen due to its flexibility, dynamic computation graph, strong research support, and efficient deployment capabilities for edge AI systems.

Database Systems

PostgreSQL – Utilized for storing structured IoT sensor data, system logs, configuration parameters, and user-related information. PostgreSQL ensures data integrity, reliability, and strong transactional support for real-time industrial applications.

MongoDB – Used for storing AI prediction outputs, image metadata, inspection results, and other unstructured or semi-structured data. MongoDB provides scalability and flexible schema design, making it suitable for handling image-based and AI-generated datasets.

API Testing and Integration Tools

Postman – Used for testing and validating FastAPI endpoints, IoT communication APIs, and AI inference services. Postman enables systematic verification of RESTful services, ensuring reliable communication between edge devices, databases, and cloud components.

4. SYSTEM ARCHITECTURE

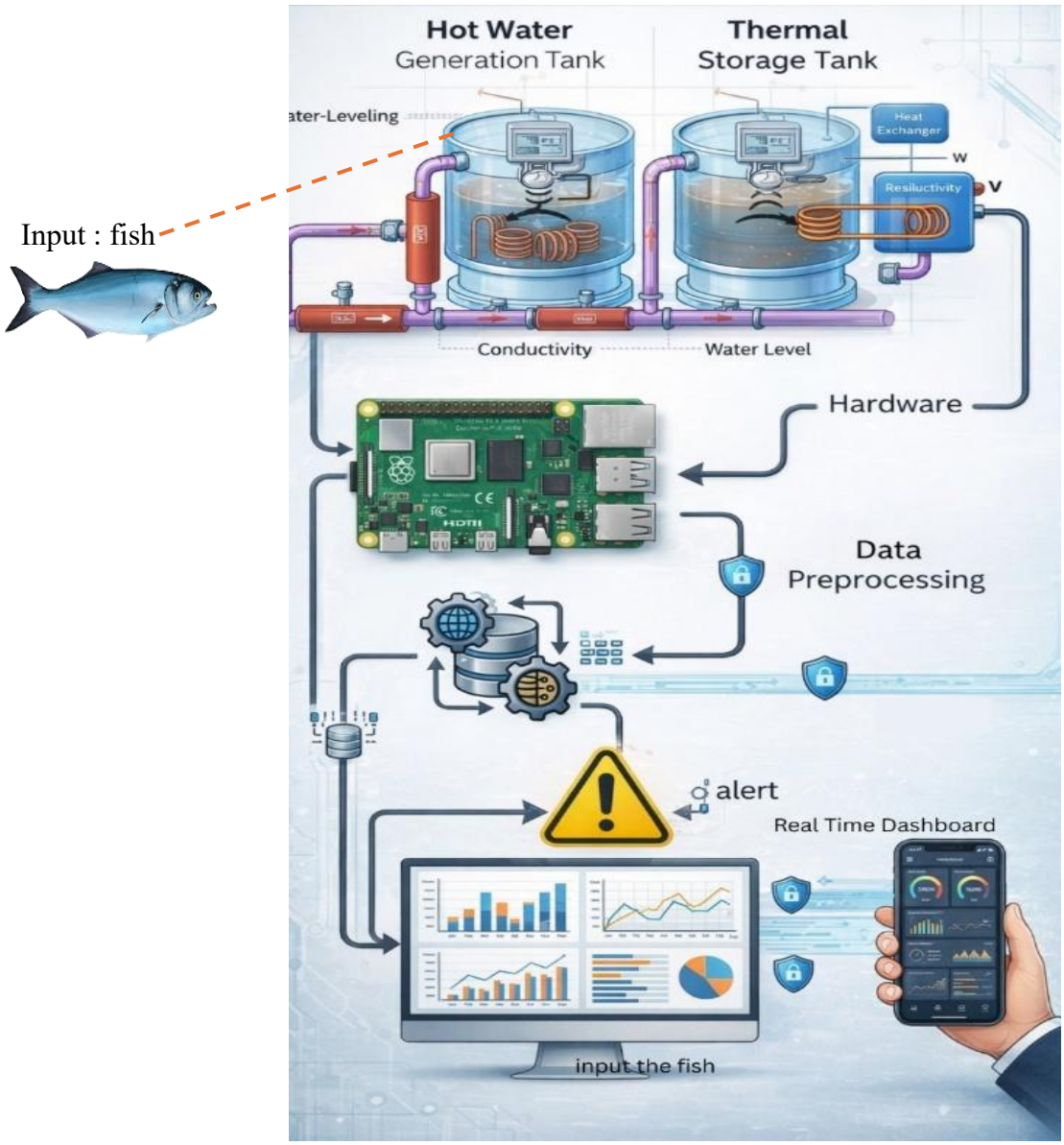


Figure 1 - SYSTEM ARCHITECTURE

Component Overview Diagram

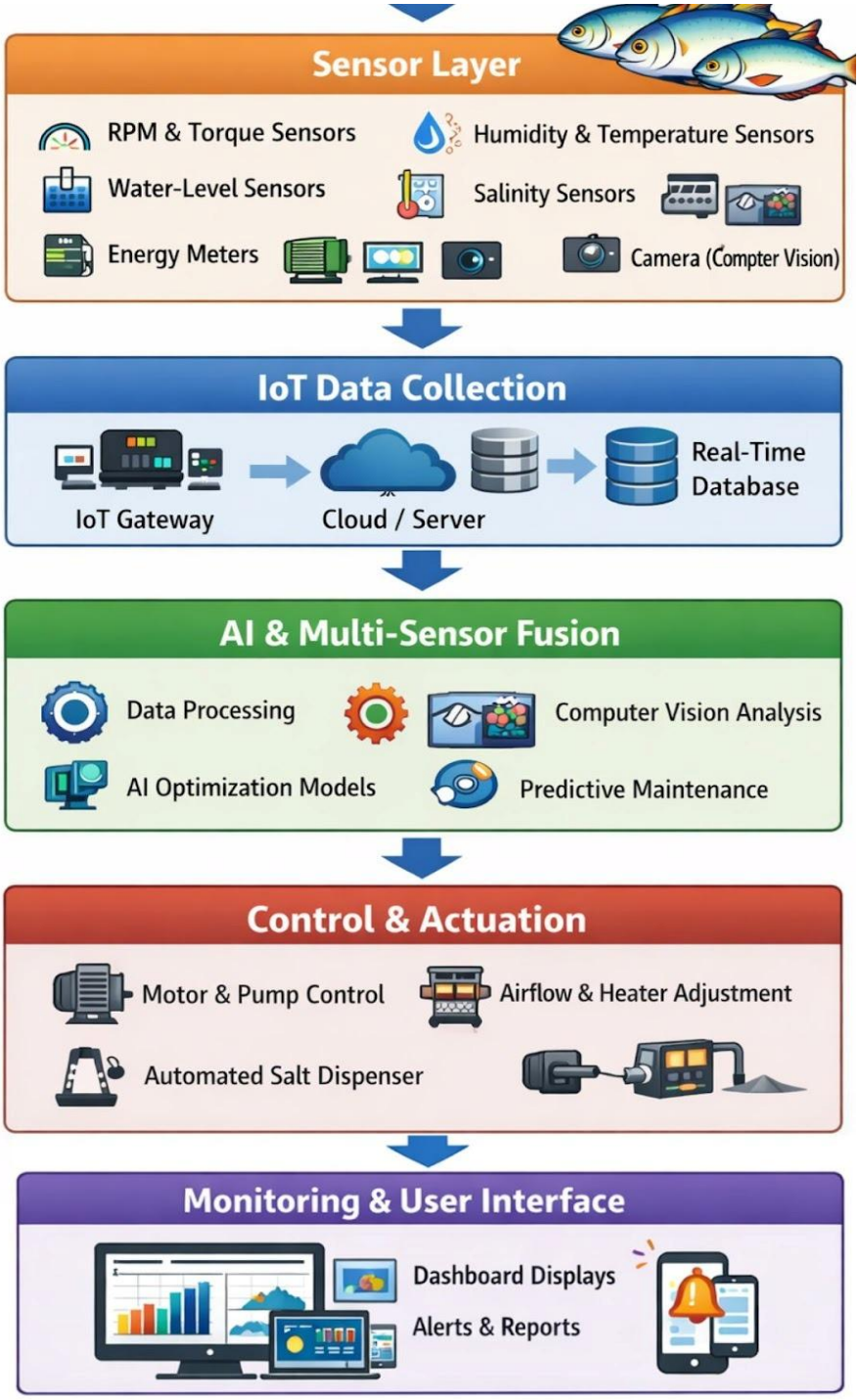


Figure 2 -Component Overview Diagram

5. USER REQUIREMENTS

Stakeholder Identification

The following stakeholders are identified as primary users and beneficiaries of the proposed system:

- **Fish Processing Operators** – Use the system for real-time monitoring and process control.
- **SME Processing Owners** – Utilize performance analytics and quality reports for decision-making.
- **Maintenance Technicians** – Monitor mechanical health indicators and predictive maintenance alerts.
- **Quality Control Personnel** – Perform product quality verification using computer vision results.
- **System Administrators** – Manage user access, system configuration, and data security.
- **Export Managers and Business Supervisors** – Analyze production performance and quality consistency reports.

Functional Requirements

The system must satisfy the following functional requirements:

1. The system shall collect real-time data from mechanical, environmental, chemical, and visual sensors.
2. The system shall monitor RPM, torque, temperature, humidity, airflow, water level, salinity, and energy consumption parameters.
3. The system shall perform multi-sensor data fusion for integrated process analysis.
4. The system shall implement predictive models to detect mechanical inefficiencies and process anomalies.
5. The system shall provide computer vision-based quality inspection for surface defect detection and salt uniformity assessment.
6. The system shall implement reinforcement learning-based adaptive control for regulating airflow, heating intensity, water level, and salinity concentration.
7. The system shall generate real-time alerts when process variables exceed predefined thresholds.

8. The system shall provide a dashboard interface for monitoring production performance and historical data trends.
9. The system shall support data storage and retrieval for process analytics and reporting.
10. The system shall enable remote monitoring and authorized user access.

Non-Functional Requirements

Performance Requirements

- The system shall process sensor data and generate control decisions within real-time operational latency limits.
- Data transmission and response time should be minimized using edge computing mechanisms.
- The system should support continuous monitoring without significant system downtime.

Security Requirements

- User authentication and role-based access control must be implemented.
- Data communication must be encrypted using secure protocols such as TLS/SSL.
- Unauthorized access to system dashboards and control modules must be prevented.

Usability Requirements

- The dashboard interface shall be user-friendly and suitable for SME-level operators.
- Visualization components shall use simple graphs, indicators, and alerts.
- System operation should require minimal technical expertise.

Reliability Requirements

- The system shall maintain stable operation under variable environmental conditions.
- Sensor redundancy mechanisms may be implemented to reduce data loss.

Scalability Requirements

- The system architecture shall support expansion to additional processing units.
- Modular hardware and software components shall allow future upgrades.

Maintainability Requirements

- The system shall allow easy sensor replacement and software updates.
- AI models should be retrainable using new production data.

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

The primary target market for the proposed system includes:

- Small- and medium-scale Maldive fish processing enterprises in Sri Lanka and the Maldives
- Commercial seafood processing factories seeking automation solutions
- Fisheries cooperative societies
- Export-oriented fish product manufacturers

Customer Persona

- SME processing owners aiming to improve production quality and efficiency
- Factory managers requiring real-time monitoring and analytics
- Quality control supervisors needing automated inspection tools

Value Proposition

The proposed system provides the following value advantages:

- Real-time monitoring of mechanical, chemical, and environmental parameters
- AI-driven quality optimization and predictive analytics
- Reduction of post-processing waste and spoilage
- Energy consumption optimization
- Automated salinity and airflow regulation
- Computer vision-based defect detection
- Scalable solution suitable for SME facilities

The system replaces experience-based manual control with intelligent data-driven automation.

Revenue Model

The commercialization strategy follows a hybrid revenue model:

1. **Hardware Sales Revenue**

    - Sensor modules, edge controllers, and actuator integration kits.
2. **Software Subscription Model**
    - Dashboard access and AI analytics platform.
3. **Installation and Maintenance Services**
    - System deployment, calibration, and technical support.
4. **Upgrade Licensing**
    - Advanced AI model updates and feature expansion.

Cost Estimation

Estimated cost components include:

- Sensor and embedded hardware procurement
- Microcontroller and communication modules
- Cloud infrastructure and storage
- Software development and testing
- Installation and calibration expenses
- Maintenance and technical support

The system is designed to be cost-optimized to ensure affordability for SME-scale processing units.

Pricing Strategy

The pricing model may follow:

- One-time hardware installation fee
- Monthly or yearly software subscription fee
- Tiered pricing based on processing capacity and feature level

Affordable pricing is prioritized to encourage adoption in developing regional markets.

Competitive Advantage

The system provides several competitive advantages:

- Domain-specific design for Maldive fish processing

- Integration of CPS, AI, and multi-sensor fusion technologies
- Reinforcement learning-based adaptive control capability
- SME-friendly scalable architecture
- Real-time quality assurance mechanism

Currently, there are limited commercial solutions specifically tailored to traditional Maldive fish processing automation.

Intellectual Property Considerations

The commercialization strategy should consider:

- Patent protection for unique AI control algorithms
- Protection of system architecture design
- Copyright protection for software code and dashboard interface
- Trade secret protection for training datasets and optimization models

Proper intellectual property management will support future technology transfer and commercial deployment.

7. BUDGET AND JUSTIFICATION

7.1 Hardware Costs

Table 2 -Hardware Budget Estimation

Item	Description	Quantity	Unit Price (LKR
Raspberry Pi 4 Model B (Common)	Edge computing device for AI processing, data aggregation, and local control	1	30000.00
ESP32 Development Board	Microcontroller for sensor interfacing and actuator control	1	2000.00
Temperature Sensors (DS18B20)	Waterproof digital temperature sensors	4	5000.00
DHT22 Temperature & Humidity Sensors	Ambient temperature and humidity monitoring	4	5000.00
Conductivity/Salinity Sensors	Analog salinity measurement probes	4	10000.00
Ultrasonic Water Level Sensors	Non-contact water level measurement	2	8000.00
Vibration Sensors (MPU6050)	Accelerometer/gyroscope for mechanical monitoring	2	11000.00
Airflow Sensors	Anemometer for drying chamber air velocity	3	3000.00
RPM Sensors	Optical/ Hall-effect speed sensors	4	6000.00
Camera Module	Camera Module	1	10000.00

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) divides the project into smaller tasks to manage development, testing, and documentation efficiently. The tasks are organized according to different phases of the research project.

8.1 Task Breakdown

Table 3 -Work Breakdown Structure (WBS)

WBS ID	Task	Description
1	Literature Review	Study research papers and identify the research gap.
2	Requirement Analysis	Identify system requirements and objectives
3	System Design	Design the overall system architecture and workflow.
4	Hardware Setup	Setup Raspberry Pi, ESP32, sensors, and camera modules.
5	Sensor Integration	Connect and test temperature, humidity, salinity, airflow, and vibration sensors.
6	AI Model Development	Develop machine learning models for prediction and optimization.
7	Computer Vision Development	Implement CNN model for fish quality and defect detection.
8	Database and Backend	Setup database and APIs for storing and managing sensor data.
9	Dashboard Development	Develop monitoring dashboard for real-time data visualization.
10	System Integration	Integrate hardware, AI models, and software components.
11	Testing and Validation	Test the system and compare results with traditional processing methods
12	Performance Evaluation	Analyze improvements in efficiency, quality, and energy usage.
13	Documentation	Prepare final report, diagrams, and research documentation.
14	Final Presentation	Prepare presentation and demonstrate the system.

8.2 Timeline Alignment

The tasks are scheduled sequentially from **research and design to implementation, testing, and final documentation** to ensure the project is completed within the allocated timeframe.

8.3 Workload Distribution

This project is conducted as an **individual undergraduate research project**, therefore all development tasks including system design, hardware implementation, AI development,

testing, and documentation are handled by the researcher with guidance from the project supervisor.

9. GANTT CHART



10. REFERENCES LIST

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11. APPENDICES

The appendices include supporting materials that provide additional details related to the proposed system design, research process, and implementation plan.

Appendix A – Survey Instruments

This appendix includes sample questionnaires or data collection forms used to gather information about current Maldive fish processing practices, operational challenges, and system requirements from industry stakeholders.

Appendix B – Detailed System Diagrams

This section contains detailed technical diagrams of the proposed system, including sensor integration layouts, IoT communication architecture, and data flow diagrams used in the system design.

Appendix C – Preliminary UI Mockups

This appendix presents early design mockups of the monitoring dashboard and user interface. These mockups illustrate how operators and system administrators will view real-time data, alerts, and performance analytics.

Appendix D – Risk Analysis

This section outlines potential technical and operational risks associated with the project, such as sensor failures, network connectivity issues, hardware malfunction, and AI model inaccuracies, along with proposed mitigation strategies.

Appendix E – Ethical Considerations and Compliance Documents

This appendix includes documentation related to ethical considerations of the research, including safe system deployment practices, responsible data usage policies, and compliance with institutional research guidelines.